Lean Agentic AI for Supplier Risk Management and Performance Evaluation

Turning spreadsheet scorecards into transparent, expert-calibrated multi-agent systems: a benchmark of four approaches and an open-source demonstrator

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Working paper, draft v0.8 · October 2026

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Abstract

Procurement organisations evaluate suppliers twice: *supplier risk management* asks how exposed the company is if a supplier fails, *supplier performance evaluation* how well it delivers. The two are often split between a licensed risk platform and simple, manually maintained spreadsheet scorecards, so one supplier is judged on two inconsistent bases; the weights behind either score are rarely checked against the judgement of the category managers who know the suppliers. This paper proposes fifteen success factors and profiles four approaches without ranking: such spreadsheet scorecards, two commercial platforms, five open-source repositories and the approach proposed here. The profiles are the author's qualitative assessment from public information and code review (October 2026), not a measurement. The paper then presents *Lean Agentic AI* and an open-source demonstrator that scores risk and performance deterministically as two views on one criteria catalogue, with deterministic agents in a LangGraph workflow, a rule-based supervisor and a human review pause. Weights are calibrated against expert labels by constrained least squares, safeguarded by shrinkage towards prior weights, a cross-validated acceptance test and human approval. An *Explain & correct* workflow shows a deterministic scoring trace whose criterion contributions add up exactly to the score; an optional language-model explanation must cite trace entries; a correction requires a reason, is re-scored in both views, shown before and after, applied only after human confirmation and recorded in a history. All experiments use synthetic data. Calibration reduced the cross-validated error from 37.8 to 20.6 points^2, close to the label-noise floor of 18.1, and the acceptance test cut harmful acceptances from up to 50 % of runs to at most 4 %. Explain & correct is verified technically, not validated with practitioners. There is no field validation and no test of predictive validity; neither fewer false alarms nor superiority over the commercial platforms is claimed.

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1 Introduction

Supplier risk evaluation has moved from an annual formality to a recurring management task. Disruptions in logistics, energy and geopolitics have made supply concentration visible, and supply-chain risk management research has long described identifying, assessing, treating and monitoring risk as one connected process [24, 11, 6]. This paper is concerned with the *assessment* stage. Due-diligence regulation expects documented risk analysis along the supply chain, and procurement teams are asked to show not only *which* suppliers are risky, but *why*, and what was done about it.

A common response is to extend the supplier scorecard: the criteria sheet gains columns for financial stability, supply-chain exposure and ESG, and a weighted average turns them into a score and a traffic light. Where budgets allow, a commercial platform supplies alerts and external data. Both responses leave a gap. A simple, manually maintained spreadsheet scorecard keeps the method in the procurement team's hands but typically keeps no record: weights drift between versions, and afterwards it is often impossible to say why a supplier was rated as it was. The platform brings data coverage and workflows, but the method by which its score is formed is, depending on the vendor, partly or largely outside the customer's view. Open-source agent projects offer modern architecture but rarely a scoring method that would stand up in an audit.

A second gap is rarely discussed. Risk platforms address *supplier risk management*; *supplier performance evaluation* — delivery reliability, quality, price, responsiveness, graded A, B or C — is a separate, established process. A common pattern in practice is that a company licenses a risk platform and still runs its supplier evaluation in spreadsheets. The same supplier is then judged in two systems, with different criteria, scales and update cycles, and nobody sees both pictures side by side. A supplier can deliver excellently and still be a critical dependency; that is exactly the case the split hides.

One problem cuts across all three options and is rarely named: **the weights are not validated.** A scorecard may weight business impact at fourteen per cent and ESG at twelve. Those numbers usually come from a workshop, a template or a vendor default, and are seldom checked against the holistic judgement of the category managers who know the suppliers best.

This paper takes a deliberately lean route. It does not propose a new risk taxonomy or data source. It starts from the scorecard practice that already exists — for risk and for performance — and asks what it would take to evaluate both on one consistent basis and to make that practice *transparent* (method and weights documented), *reproducible* (every number recomputable), *governed* (human approval where judgement is required), *explainable and correctable per result* (why one supplier got its class, and a reasoned correction with a visible effect) and *able to learn* (weights aligned with expert judgement) — at the cost profile of open-source software. The argument proceeds in two steps: a benchmark of four approaches on fifteen success factors (Figure 1), read factor by factor without weighting, total or ranking, as a qualitative expert assessment by the author rather than a measurement; and an open-source demonstrator, so that each claimed property can be inspected in code, tests and documentation rather than asserted. The demonstrator and all experiments run on synthetic data; no real expert labels or field data were used. This has a cost, discussed in Section 10, and a benefit: because the labels were generated from a known hidden weighting, it can be checked whether calibration recovers it.

Terminology. "Agentic" denotes an *orchestrated multi-agent workflow*: the agents (data collection, scoring, calibration, recommendation, human review) are deterministic Python functions acting as nodes of a LangGraph state graph, a rule-based supervisor routes between them, and the workflow pauses for human approval. Scoring, calibration, recommendations and routing use no language model. The only exception is an optional *Supplier Review Agent* that, on explicit request, asks a language model to explain one supplier's deterministic scoring trace and cannot change scores, classes or weights (Sections 6.9 and 6.10); document extraction and an LLM-based supervisor are planned extensions only. The term does not imply autonomous, LLM-driven decision making.

Contributions.

1. **Fifteen success factors** for supplier risk management and performance evaluation, including their integration, each with a definition and the evidence by which it can be assessed, together with the documented reviews that produced the current version (Section 3).

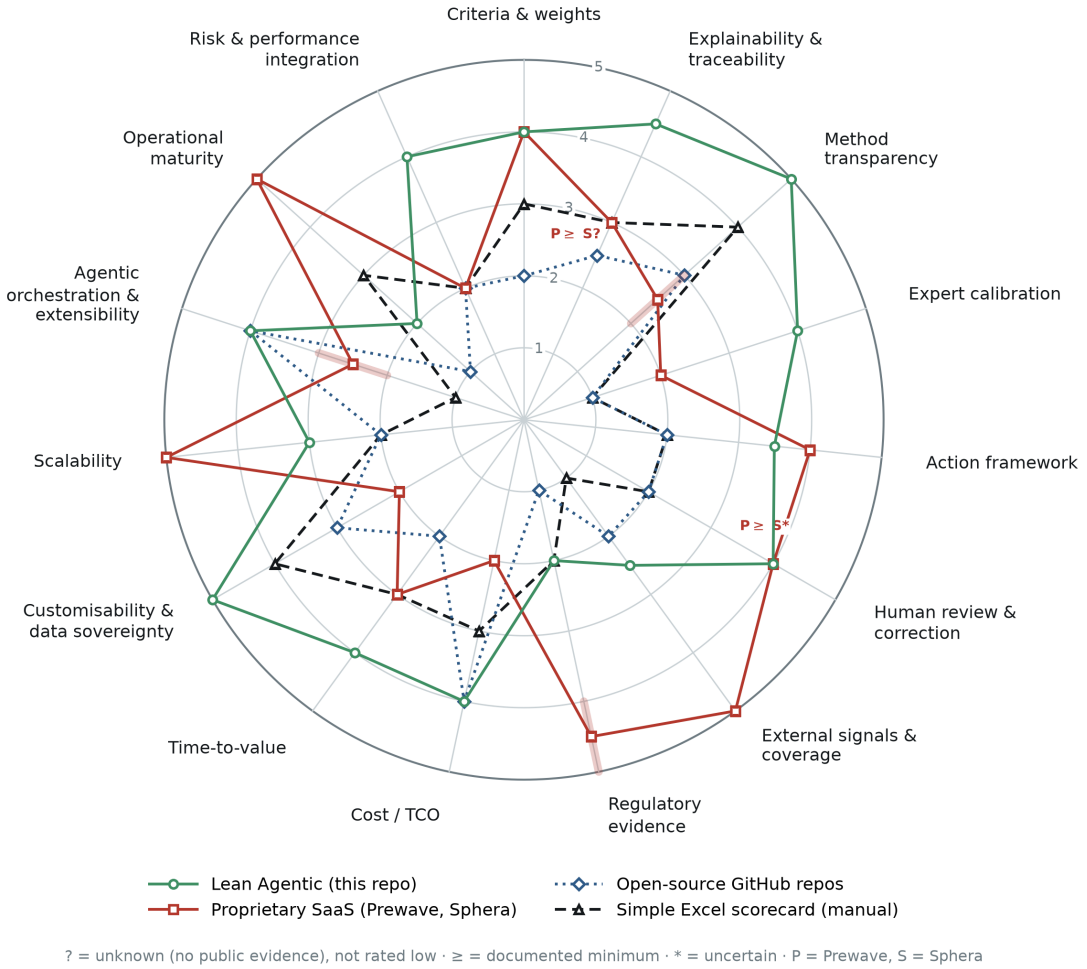


Figure 1. Four approaches to supplier risk management and performance evaluation profiled on fifteen success factors (model v0.5), on a five-level scale; half points mark a profile between two levels. Proprietary SaaS is plotted as the mean of the known Prewave and Sphera values; shaded bars mark where the two differ. Unknown values (no public evidence) are not plotted as low: "?" marks an unknown value, "S?" a point where Sphera is unknown and only Prewave is shown, "≥" a documented minimum (P = Prewave) and "S*" a Sphera value kept but uncertain. Rationale and confidence for every value: Table 12 and `research/comparison-matrix.md`. **Illustrative, qualitative assessment by the author based on public information and code review, October 2026; expert judgement, not an empirical measurement.** The author also built one of the rated approaches and wrote the v0.5 rating anchors after building the feature they reward (Section 4.4). The factors stay separate: no weighting is applied, no total is formed and no ranking is derived.

2. **A profile benchmark** of four approaches. Every value comes from one machine-readable source with a rationale and a confidence mark; the two commercial examples are rated separately (Section 4).
3. **The Lean Agentic approach**, a synthesis of scorecard practice, platform expectations and open agentic architecture. Its positioning rests on two features: the integration of risk management and performance evaluation, and calibration against expert labels inside an agentic workflow (Section 5).
4. **An open-source demonstrator**: a LangGraph workflow of deterministic agents with deterministic scoring, material-group weights, constrained least-squares calibration, a graph-interrupt review step, versioned weights, two views (performance evaluation and risk management), an action framework, an assessment history and an optional, experimental LLM-based review of final results that changes no number (Section 6), examined in five experiments on synthetic data (Section 7).
5. **Explain & correct**, which makes each result explainable and correctable (Sections 6.10 and 7.9): a deterministic scoring trace whose criterion contributions add up exactly to the score; an optional language-model explanation that must cite trace entries; corrections of data or ratings with a mandatory reason, kept apart from expert overrides; deterministic recomputation of both views with a before/after comparison; and human confirmation with a correction history. It is verified technically, not validated in the domain.
6. **An explicit statement of what is not established**, and of the work that would close each gap (Sections 10–11).

2 Background and related work

2.1 Supplier risk assessment in practice

Weighted multi-criteria aggregation is the dominant instrument for evaluating suppliers. Reviews of the supplier-selection literature catalogue the analytic hierarchy process, outranking, data envelopment analysis, mathematical programming and fuzzy or hybrid approaches [10, 4]; most share one structure of criteria, weights and an aggregation rule. Case studies show formal and informal supply-risk assessment processes in use, frequently in scoring form [26]. A typical supplier-risk scorecard has a *severity* component (how replaceable is the supplier, how badly would a failure hurt) and *likelihood* components for financial stability, supply-chain exposure and ESG or compliance, rated on an ordinal 1–5 scale, weighted and aggregated into a score and a class. The method is not in question here; how it is run is: definitions are informal, weights are set once, results are not reproducible after the file has changed, and consequences are free text, if they are written at all.

2.2 Commercial supply-chain risk platforms

Two platforms serve as named examples, assessed **on public information only** (vendor websites, help-centre articles, press releases, as of October 2026); nothing here is a product test. **Prewave** combines continuous media and trade-data monitoring with supplier assessments and due-diligence modules. Unusually, it publishes the composition of its supplier score: a peer component (country, industry, commodity) at 30 %, alerts at 40 %, assessments at 20 % and external data at 10 %, aggregated non-linearly. Event types and weights can be adjusted in customer-specific "perspectives"; source weighting and signal classification are not disclosed. **Sphera Supply Chain Risk Management** offers portfolio risk views, multi-tier mapping, early warnings, response playbooks and a documented bundle for the German supply-chain due-diligence act. Its scores are described as "AI-scoring" tied to business impact; no public description of the method or of customer-adjustable weights was found.

Both use AI: Prewave for signal processing and with connectors for external AI agents, Sphera with fine-tuned generative models for supplier risk summaries and marketed "agentic AI" capabilities. Three observations matter. "Platform" is not one methodological profile: on transparency the two examples differ by a full scale point, which is why they are rated separately. **Neither documents calibration against the customer's own expert judgement**: weight adjustment exists, but as configuration, not as a fitted and reported procedure. And **both are risk platforms**: neither documents a supplier performance evaluation with its own criteria and grades, nor a customer-inspectable multi-agent workflow — specialised agents over a shared state, a calibration agent, a review pause in the workflow graph — in public sources. This describes public documentation as of October 2026, not the products themselves.

2.3 Open-source agentic supply-chain projects

Five public repositories were reviewed at source-code level. They converge on one architecture — specialised agents sharing a typed state, coordinated by a graph or a supervisor — and diverge sharply on scoring. One contains no scoring at all; two compute deterministic, hard-coded weighted scores, one of them deliberately overwriting numbers produced by the language model; two leave the risk judgement to a language model, and in one a published example contains arithmetic errors in the weighted totals, which illustrates the risk of letting a language model generate numbers. One carries no licence, two state MIT only in their README without a licence file, one is under a copyleft licence (GPL-3.0), and none ships an automated test suite. The repositories are most useful here as **architectural blueprints**; their focus differs from the scoring-method focus of this paper. This work reuses ideas from them, but no code.

2.4 Calibrating weights against expert judgement

Two practices both go by the name of "setting weights": *eliciting* them directly by asking how important each criterion is, and *inferring* them from holistic judgements of how risky each supplier is overall.

Direct elicitation is method-dependent. Different procedures (swing, SMART, trade-off, pairwise comparison) produce systematically different weights for the same problem [3, 22], and weight judgements are subject to range insensitivity and splitting biases [25]. Weights agreed in a workshop are one plausible answer among several, not a measurement.

Inferring weights has two traditions. *Judgement analysis and policy capturing*, rooted in Brunswik's lens model [8], regresses holistic judgements on cues to recover the weights a judge implicitly uses [1, 16]. Linear models reproduce such judgements well [15], and a model of a judge can outperform the judge by applying the policy consistently [7]. The tradition also cautions that "improper" weights, for example equal weights, often perform about as well as fitted ones [5]: gains from fitting should be modest, and fitted weights should not stray far from sensible priors. *Preference disaggregation* infers an additive value model from holistic evaluations, in the UTA family by linear programming on rankings [13, 14]. The calibration here belongs to the same family but fits *cardinal* scores by least squares, constrains the weights to a simplex with a lower bound, and shrinks them towards a prior. With few judgements and many weights, a fit adapts to noise; ridge regularisation trades a small bias for a large reduction in variance [12], and shrinking towards a prior instead of zero is a direct generalisation [9]. Directly elicited weights

thus serve as the prior, and holistic labels adjust them under constraints.

2.5 Language models and determinism

For risk scores, language models are the wrong default. A score that changes when the same input is presented twice cannot be audited; outputs can vary across identical requests even at temperature zero, as shown for code generation [21]. A score that cannot be recomputed by hand cannot be challenged. The more mature reviewed designs compute numbers in code and reserve the language model for language. This paper adopts that division strictly: its one language-model component (Section 6.9) only interprets numbers that code has already computed.

3 Success factors

An approach is easier to assess when the properties that matter are named before any approach is rated. Table 1 proposes fifteen such *success factors*, each with the observable evidence that would let someone else check a rating. They form a **flat list**: not arranged in a hierarchy, not weighted and not combined into a total. An approach is read as a **profile**; one that is strong on twelve factors and absent on one is a different proposition from one that is uniformly adequate, and a single figure would treat the two alike.

3.1 How the factor model was derived, and corrected

The first version (v0.1) was drawn up from the requirements of the proposed approach. A critical review identified four problems, documented because they bear on how far the benchmark can be trusted. *Means presented as ends*: "open source" and "multi-agent orchestration" named the means one had chosen, which makes the comparison circular; both were reframed as outcomes (customisability and data sovereignty; extensibility). *Home-field bias*: v0.1 contained only one factor on which platforms are clearly stronger; *regulatory evidence* and *operational maturity* were added. *Overlaps*: structured criteria and objective/subjective inputs, and audit trail and determinism, were merged; transparency (understanding the method before use) was kept apart from traceability (reconstructing a result afterwards). *Validity*: no factor asks whether a score *predicts* disruptions. No reviewed approach reports predictive validity publicly, so the factor cannot be rated today; it is left off the radar and treated as an open question.

A second review (v0.3) followed informal practitioner feedback, not a structured survey (numbers in this paragraph refer to v0.3). It added factor 16, *integration of risk management and performance evaluation*, because the common split between a risk platform and spreadsheet-based evaluation was not captured by any factor. It sharpened factor 14 to *agentic orchestration and extensibility*, because "uses AI" and "is an agentic workflow" were indistinguishable; the factor stays an outcome — explicit, inspectable, extensible orchestration — but its evidence now names agents with shared state and review pauses. And it widened factor 10 to the total cost of ownership, including recurring manual effort. Factor 16 is one on which the proposed approach is strong, and factor 14 moves partly back towards an architecture property that v0.2 had reframed; both are disclosed as such. The four factors on which the commercial examples are clearly ahead (8, 9, 13 and 15) are unchanged.

A third review (v0.4) reduces the model to fifteen factors. Material-group adaptation (old 5) becomes evidence under customisability and data sovereignty (old 12, now 11): per-commodity configuration is a specific case of changing the method. Its separate axis double-counted that capability. Integration remains unchanged (old 16, now 15); old 1–4 retain their numbers and old 6–16 become 5–15. All retained scores, confidence marks and per-vendor values stay unchanged. The customisability rationales now include group configuration and its limits; the removed scores are retired, not averaged. The retained customer-control ratings are Excel 4/M, SaaS 2/M, open repos 3/H and Lean Agentic 5/H. For open repos, visible code supports modification, but neither supplies group adaptation nor removes licence constraints. Platform strengths retain their scores (now 7, 8, 12, 14).

The remaining boundaries separate structure from control, configuration from fitted calibration, method transparency from historical reconstruction, actions from approval, lifecycle cost from adoption time and capacity, customer control from workflow extension, and generic audit records from regulatory documentation. Governance concerns decision rights; operations concerns reliable service, infrastructure security and support. Joint risk/performance evaluation remains distinct from a structured single-view model. This is conceptual separation, not proven statistical independence: a 4×15 rating matrix has rank at most four, or three after centring. Independent validation is still needed; the full overlap review is in `research/success-factors.md`.

A fourth review (v0.5) keeps fifteen factors and sharpens two. Factor 2 becomes *explainability and traceability (per result)*: can a user see why one supplier got its score and class — inputs, sources and data dates, missing data, rules, weights, contributions, non-additive rules, explanations citing them — and reproduce it? Factor 6 becomes *human review and correction*: can a person review, dispute, correct data or override a rating with a reason, see the effect before confirming, and find the decision again? Levels 1–5 are anchored in writing for factors 2, 3 and 6. Boundaries keep one capability from being rewarded twice: 3 is the method before use, 2 the reasons for one result; 2 is what the system records and explains, 6 what people decide and change; 13 is the review pause as architecture, 6 what a reviewer can do in it; a language-model explanation counts under 2 only. *Unknown is not absent*: a value without public evidence

Table 1. Success factors for supplier risk management and performance evaluation (model v0.5). The factors are read separately; no weighting, total or ranking is derived from them.

	Success factor	What it means	Assessable by
1	Structured criteria & weights	Explicit criteria, scales and weights combining measurable indicators and structured expert assessment	Criteria with definition and scale; explicit weights; both input types with provenance
2	Explainability & traceability (per result)	For one supplier, a user can see why it got its score and class, and the result can be reproduced	Inputs with source and data date; missing data marked; rules, weights and contributions per criterion; non-additive rules separately; per-result record; recomputation; explanations citing the trace
3	Methodological transparency	The method is documented and checkable *before* use	Published formula, weights, sources, rules
4	Expert calibration	Parameters are systematically fitted to expert judgement, with approval	Procedure; fit metric; logged old and new parameters; approval
5	Action framework	Results map to defined actions, review cycles and owners	Actions per class and driver; tracking to closure
6	Human review & correction	People can review, challenge and correct results in a controlled way; decisions are recorded	Review and approval points; corrections and overrides with mandatory reason; before/after effect before confirmation; dispute channel; decision history; roles
7	External signals & coverage	Current external data and breadth of supplier coverage	Sources, update frequency, languages, tier-N
8	Regulatory evidence	Support for due-diligence documentation duties	Regulation-specific workflows and reports
9	Cost / TCO	Total cost of use, including recurring internal effort, and access barriers	Licence model, prices, implementation effort, manual effort per cycle
10	Time-to-value	Time and effort to the first usable result	Time to first portfolio assessment
11	Customisability & data sovereignty	The organisation can change method and implementation, including material-group configuration, and controls its data	Configurable without vendor; per-group criteria, weights or thresholds; licence; data location
12	Scalability	Works as suppliers, users, groups and update frequency grow	Suppliers handled; marginal effort; automation
13	Agentic orchestration & extensibility	Work is orchestrated as an explicit workflow of specialised agents; new capabilities can be added without rebuilding the core	Specialised agents over a shared, inspectable state; review pauses in the workflow; extension points; APIs
14	Operational maturity & support	Production readiness	Support, service levels, security, release process
15	Risk & performance integration	Supplier risk and supplier performance are evaluated on one consistent basis, each with its own logic	Shared criteria and data; separate weights, classes and actions per view; both results per supplier

is marked unknown and not plotted as low, and one covering only the documented part is marked as a documented minimum ($\geq$); a value kept at its earlier level because the evidence neither confirms nor contradicts the new anchor is marked uncertain (*). Re-rated values: factor 2, Excel 3 (was 2), SaaS 3 (Prewave $\geq$ 3, Sphera unknown; was 2.5), open repos 2.5 (was 2), Lean Agentic 4.5 (was 4); factor 6, Excel 2, SaaS 4 (Prewave $\geq$ 4, Sphera 4 uncertain; unchanged), open repos 2, Lean Agentic 4 (was 3.5). For Sphera the sources document level 3 of the new anchor; the level-4 element "corrections or overrides with a reason" was not found, and since nothing says it is absent, the earlier value is kept and marked uncertain rather than lowered on missing evidence. Platform sources were re-checked on 12 October 2026 (Prewave help centre: score components with weights, alert reporting with reasons reviewed by Prewave, supplier portal; Sphera: AI summaries at marketing level only), the repositories in code. v0.5 and its anchors were written by the author *after* building the Explain & correct workflow they reward (Section 4.4).

4 Benchmark: four approaches profiled

4.1 Method

Four approaches are rated on a five-point scale (1 weak or absent, 3 adequate, 5 very strong); a half point marks a profile clearly between two levels. The ratings are the author's qualitative expert judgement

from public information, source-code review and practitioner experience; they are not measurements, and no product was tested. Each rating carries a rationale and a confidence mark: H verified from code or primary documentation, M plausible and partly documented, L assumption. A value without public evidence is marked *unknown* rather than rated low, one covering only the publicly documented part as a documented minimum ($\geq$), and one kept at its earlier level because the evidence neither confirms nor contradicts the anchor as uncertain (*); no value is lowered only because evidence is missing; the SaaS profile is the mean of the known Prewave and Sphera values. All ratings live in one machine-readable file from which Figure 1 and Table 12 are generated; a consistency check fails if the published table no longer matches the source. The evidence differs by approach, and this is disclosed (Table 2): spreadsheet practice is a stereotype of a simple, manually maintained scorecard, based on the author's practitioner experience; Prewave and Sphera are rated separately on public information; the five repositories by source-code review; and the demonstrator *as implemented*, not its roadmap.

4.2 Reading the profiles

The simple spreadsheet scorecard is strong where simplicity helps: it keeps formulas visible and can be changed by the team itself. It costs no licence but a great deal of recurring manual work, and a governed scorecard takes weeks to build. As rated, it typically lacks the system properties — an audit trail, learning, external data, orchestration — and risk and performance usually live in separate files. This is a statement about the rated profile, not about spreadsheets in general: real practices vary widely, and a well-governed workbook can do better than the stereotype. **The commercial platforms are strongest on data, scale and operations**: external signals, regulatory evidence, scalability and operational maturity, with well-developed workflows. They are weaker on cost, self-determined customisation, recomputability by the customer (a design consequence of continuous monitoring rather than a flaw) and, for one of the two, transparency; their spread is largest exactly there. On explanation per result and human review the public evidence is incomplete: Prewave documents score components with weights and alert reporting with reasons (documented minimums), while for Sphera no per-result breakdown was found, so that value is unknown rather than low. They are risk platforms: performance evaluation is not part of their documented scope, and their AI is not exposed as an inspectable multi-agent workflow. **The open-source repositories are blueprints**: free to read but not all free to reuse, and demo-grade, with agent architectures as their main asset; they are rated rudimentary on most methodological factors, which reflects a different focus rather than a failed method. **The Lean Agentic demonstrator is strongest where the method itself matters**: transparency, explainability and traceability of each result (4.5, through the scoring trace of Section 6.10), expert calibration and customisability (including material-group configuration) — and on the two factors that set it apart from pure risk platforms, the integration of risk management and performance evaluation and agentic orchestration (both 4, against 2 and 2.5 for the platforms). On human review and correction it is rated 4, the same as the platforms (Prewave a documented minimum, Sphera uncertain): reasoned corrections, a before/after preview and a history exist, but roles and verified identities do not, where the platforms describe ownership and approval workflows. On actions it is close to the platforms (3.5 against 4); tracking to closure is missing. It is better than the rated spreadsheet scorecard on cost of ownership and time-to-value, because scoring, actions and reports run automatically from a template. On external signals it reaches about half of the strongest platform (2.5 against 5): the provider interface reads open data and system exports and records their origin, while live APIs are still placeholders and there is no continuous monitoring. The architecture allows further providers, and an external-

Table 2. Condensed benchmark (author's assessment, not a measurement): strongest (4–5) and weakest (1–2) factors per approach; factors rated 1 are in bold. Full matrix: Appendix, Table 12.

Approach	Strongest	Weakest
Excel / manual	Transparency; customisability	**Calibration; external signals; agentic orchestration**; actions; human review & correction; regulatory evidence; scalability; risk & performance integration
SaaS (Prewave, Sphera)	External signals; scalability; operational maturity; regulatory evidence; criteria & weights; action framework	Cost; customisability; calibration; risk & performance integration
Open-source repos	Cost; agentic orchestration	**Calibration; regulatory evidence; operational maturity**; most other factors
Lean Agentic	Transparency; customisability; explainability & traceability; risk & performance integration; calibration; agentic orchestration; human review & correction; criteria & weights; cost; time-to-value	Regulatory evidence; operational maturity

signal agent is a natural extension; this upside is not rated. The approach remains weak on regulatory evidence and operational maturity. These dips define what the approach is not.

4.3 Complementary, not ranked

The profiles do not order the approaches. Lean Agentic does not replace the data coverage of a platform. The realistic reading is **complementary**: a platform or open data serves as the *signal source*, and a transparent, calibrated model owned by procurement turns signals and internal knowledge into classes and actions (Section 8.2).

4.4 Threats to validity

Author bias: the factors and all ratings come from a single rater, the author, who also built one of the rated approaches and thus rates their own work. The v0.2 correction, rating the demonstrator as implemented, and a rationale and confidence mark for every value reduce the bias but do not remove it. The v0.3 review works in the other direction: it added a factor on which the proposed approach is strong and re-rated five factors after practitioner feedback, all of them in its favour (action framework, human-in-the-loop, time-to-value and external signals upwards; cost of the spreadsheet scorecard and the repositories downwards). The re-ratings are documented with their rationale and rate the implementation, not the roadmap; expected upside, for example in signal processing, is named but not rated. The v0.5 review also favours the proposed approach: it sharpened factors 2 and 6 and added rating anchors after the author had built the Explain & correct workflow, so the anchors were written by the person who built the feature they reward and fit its design. They make the ratings checkable against a written standard; they do not make them independent. An independent re-rating against the published anchors is future work. *Public information only* for the platforms: hands-on evaluation could move several ratings; values marked unknown or $\geq$ are the most likely to change. *Date dependency*: all platform ratings refer to October 2026. *Ordinal scale*: differences of one point are not measured distances.

5 The Lean Agentic approach

5.1 Definition and synthesis

Lean Agentic AI is defined here as an approach that starts from the evaluation practice an organisation already has, uses open agentic workflows as blueprints, and builds a tailored, transparent, low-cost system on top of that practice rather than replacing it. "Lean" means building only what adds value, reusing open blueprints, and keeping the method with the team that is accountable for it.

Positioning: one basis for risk and performance. The approach is not a further risk platform. Its main difference from pure risk platforms such as Prewave and Sphera, as far as their public documentation shows (October 2026), is that it evaluates *supplier risk management* and *supplier performance evaluation* on one consistent basis: one criteria catalogue, one data sheet, one engine, and two views with their own logic (Section 6.6). This closes the gap described in the introduction, where a risk platform and a spreadsheet evaluation judge the same supplier separately. It does not replace the platform's signals; it can consume them as indicators (Section 8.2). The second pillar is that the method is run by an agentic workflow whose agents — including a calibration agent — share an inspectable state and stop for human approval inside the graph. The agents that score, calibrate, recommend and route are deterministic code without a language model; an optional review agent uses one only to interpret final results (Section 6.9).

The approach inherits from each tradition in Section 2. *From scorecard practice* it takes the domain logic: named criteria, 1–5 ratings, weighted dimensions including a severity component (availability of alternatives, change complexity, business consequence) and class-based actions. Procurement keeps its method and vocabulary; the method becomes explicit, versioned and testable. *From commercial platforms* it takes the expectations placed on a professional system: review and approval steps, an audit trail, integration of objective indicators with expert assessment, and an action framework. External signal feeds are not implemented; they are intended to come from open data or a platform, through the provider interface of Section 6.7. *From open-source agent projects* it takes the architecture — a supervisor coordinating specialised agents over a typed state — and the lesson that numbers are computed in code.

5.2 Distinguishing features

Nine features distinguish the approach; the first three carry its positioning. **(1) Integration of risk management and performance evaluation.** Each criterion is defined once, with its meaning and rating source, and shared where both views need it (for example credit risk or compliance status). Each view has its own dimensions, weights, classes, escalation rules, actions, expert labels and approvals; risk is classed low to critical, performance graded A to C. A supplier is seen in both views from the same data, and calibrating one view never changes the other. Neither commercial example documents performance evaluation as part of its platform. **(2) Calibration against expert judgement**, the central methodological contribution. Each labelled supplier receives a holistic risk score from 0 to 100 on the model's own scale, and the weights are chosen to minimise the squared error between model score and label, subject to a minimum weight per factor and weights summing to one. A ridge term pulls the solution towards the configured defaults, which protects small samples from overfitting and keeps criteria the methodology considers relevant from being switched off silently. The

anchor is always the configured default, never a previous calibration, so repeated calibration on the same labels cannot drift; this problem was found by a test. Weights are accepted only if the *cross-validated* error falls clearly (Sections 7.6 and 7.8) and are then applied or proposed for approval, as configured (both bundled templates propose by default), and old and new weights, errors and class agreement are reported. None of the reviewed repositories, and neither commercial example in its public documentation (as of October 2026), offers this. **(3) Agentic orchestration.** The method runs as a graph of specialised agents — data collection, scoring, calibration, recommendation, human review — over one typed, inspectable state, coordinated by a rule-based supervisor with bounded loops; each of these agents is a deterministic Python function without a language model, and only the optional review agent calls one, to interpret final results. The calibration agent proposes weights; the review is a pause in the graph that resumes with a recorded decision. The commercial examples use AI for signal processing, summaries and, in one case, marketed "agentic" capabilities, but neither documents such a workflow publicly. The agents orchestrate; they do not decide (Section 8). **(4) Weights as versioned objects**: default, material-group and calibrated weights are interchangeable, never hard-coded, and every applied set is written as a new version with timestamp, approver, model version and metrics. **(5) Material-group awareness**: weights can be set and calibrated per commodity, and each supplier is scored with the most specific set available. **(6) Human-in-the-loop as a real pause**: when data is insufficient or weights are proposed, the workflow stops until a reviewer approves weights, sets a class with a comment, or defers; manual classes survive re-scoring. **(7) Deterministic scoring**: every number is produced by pure, tested functions and can be recomputed by hand; language models are kept out of the numbers, and their one implemented role is an optional interpretation of final results. **(8) Governance by construction**: escalation rules set class floors that no average can hide, a completeness check flags results resting on too little data, the audit trail is append-only, and every feedback loop is bounded. **(9) Explain & correct**: a deterministic trace lists per criterion the inputs with source, the rating rule, the effective weight and the contribution, which add up exactly to the score; non-additive rules are explained separately. A correction needs a reason, is re-scored in both views, shown as before/after and applied only after confirmation. Weights, thresholds and rules cannot be changed this way (Section 6.10).

5.3 What the approach is not

It is not a data provider (it monitors no news, disasters or tier-N networks, and ships no real external data), not a regulatory compliance product, not production-hardened (no roles, no central run store), and not empirically validated (calibration has been shown on synthetic labels only). The claim is narrower: the *method layer* of supplier risk evaluation can be made transparent, reproducible, governed and able to learn at low cost — a layer that platforms leave partly opaque and that simple, manually maintained spreadsheet scorecards typically leave ungoverned.

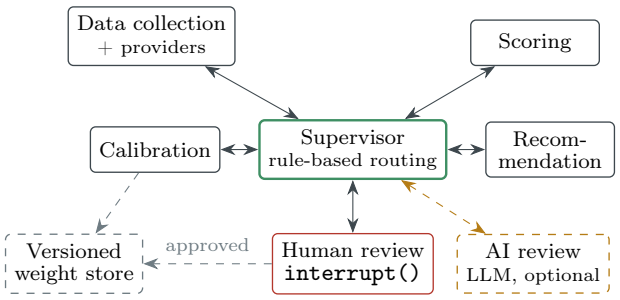


Figure 2. Workflow of the demonstrator. All nodes that score, calibrate, recommend and route are deterministic and call no language model. The dashed AI review node is entered only on an explicit request and interprets final results without changing them (Section 6.9). Every worker returns to the supervisor; calibration (with its rescore) and human review each run at most once per run. Accepted weights go to the versioned store only if auto-applied or approved.

6 Architecture of the demonstrator

6.1 Workflow

The demonstrator is a LangGraph state graph in hub-and-spoke form (Figure 2). A rule-based **supervisor** inspects the shared state and routes to one of five workers, plus an optional sixth: *data collection* loads the suppliers in scope (optionally one material group) with indicators and expert labels, and lets optional data providers fill missing indicators; *scoring* rates factors, aggregates, normalises, classifies, applies escalation rules and checks completeness; *calibration* is optional; *recommendation* maps classes, elevated dimensions and factor levels to actions, owners and review cycles; *human review* pauses the run; the optional *AI review* runs only on explicit request (Section 6.9). Every worker returns to the supervisor, and the calibration loop and review loop each run at most once per run. The supervisor and the five core workers are deterministic Python code without a language model; the AI review node is the only place that calls one. The routing rules fit in one function, which is also the single, so far unused, extension point for a planned language-model-based supervisor.

6.2 Scoring model

Each factor is rated on an ordinal scale from 1 (low risk) to 5 (high risk), either by an expert or by a rule that converts a raw indicator; an expert rating always takes precedence, and the provenance of every rating is recorded. With effective factor weights θ (non-negative, summing to one), dimension ratings

are weighted means of their available factors, and the overall rating R is the weighted mean of the dimensions. R is mapped to a score by min-max normalisation,

$$S = \frac{R-1}{4} \cdot 100, \tag{1}$$

so that the best possible supplier scores 0, the midpoint 50 and the worst 100; a naive $R/5$ would start at 20 and waste a fifth of the range. The mapping is affine, so rankings are unchanged and calibration stays linear. Classes follow from thresholds on S (in the template: low < 30, medium < 50, high < 70, critical from 70). Escalation rules can raise a class but never lower it. Missing factors are excluded and the remaining weights re-normalised; the share of weight backed by data is reported and, below a threshold, flagged for review.

Sub-scores. Dimension scores are reported alongside the overall score. Two combine data from different origins. **Supply chain risk** combines *political stability*, from the percentile rank of the World Bank governance indicator "Political Stability and Absence of Violence/Terrorism" [17] in bands of 20 points ($\geq 80 \to 1$, …, $< 20 \to 5$, with a generic country risk index as fallback), *sourcing strategy* (number of qualified sources) and *transport dependency* (mode and routes). **Severity** includes *change complexity*: how long it would take to move the volume to another source, including qualification, tooling and ramp-up. It can be derived from the *time to exit* in months (≤ 3, ≤ 6, ≤ 12, ≤ 24, > 24 months $\to 1, \ldots, 5$); an expert rating still takes precedence. Both mappings are ordinary rating sources with illustrative, uncalibrated bands; only weights are calibrated. The default criteria, weights and thresholds are an **illustrative template**, not a recommendation.

6.3 Weights as objects

The engine never reads weights from code. Each supplier receives the most specific available weight set: calibrated weights for its material group, configured weights for that group, calibrated global weights, or the configured defaults. Applied sets are persisted in a versioned store and loaded at the start of the next run.

6.4 Calibration

For n suppliers with complete ratings $\mathbf{x}_i$ and labels $y_i \in [0, 100]$, the calibrated weights solve

$$\begin{aligned} \min_{\theta} \ & \frac{1}{n} \sum_i \Big(\frac{s_i(\theta) - y_i}{100} \Big)^2 + \lambda \, \|\theta - \theta_{\text{anchor}}\|^2 \\ \text{s.t.} \quad & \theta_f \geq w_{\min} \ \ \forall f, \qquad \textstyle\sum_f \theta_f = 1, \end{aligned} \tag{2}$$

with $s_i(\theta) = (\mathbf{x}_i \cdot \theta - 1)/4 \cdot 100$ and θ_{anchor} the configured defaults. This is a convex quadratic program, least squares with linear constraints [19], solved by sequential least squares with an analytic gradient. Defaults: $\lambda = 0.05 \times 32/n$ (Section 7.4), $w_{\min} = 0.02$, at least eight labels with complete data, and acceptance only if the cross-validated error falls by at least 5 % (5 folds × 3 repeats) *and* by at least two standard errors ($z \geq 2$; Section 7.8). Only complete suppliers are used, since missing-data re-weighting would make the score non-linear in θ. The report also checks for a **level offset** between labels and scores, which weights cannot express (Section 7.7); a flagged offset withholds the calibration for review. Labels are expected as scores rather than classes, which carry less information and would make the fit non-linear; class agreement is reported alongside the error.

6.5 Human review

The review step uses the graph framework's interrupt mechanism. The run is checkpointed, and a structured request lists flagged suppliers with reasons and any proposed weights with their metrics. The run resumes with a decision object: approved weights are stored with the reviewer's name and the portfolio is re-scored; manual classes survive re-scoring and are marked as overrides; recommendations are rebuilt after any change; invalid decisions are reported and ignored.

6.6 Two views: performance evaluation and risk management

Practice distinguishes two questions about the same supplier. **Performance evaluation** looks back — how well does the supplier deliver? — at delivery reliability, quality, price and responsiveness, and grades suppliers A, B or C. **Risk management** looks forward — how exposed are we if the supplier fails? — at replaceability, financial stability, supply-chain structure and compliance. The answers can diverge: in the sample portfolio, two suppliers are critical in the risk view and A in the performance view. The demonstrator therefore keeps both as separate **views** on one engine; this integration is the approach's main difference from pure risk platforms as publicly documented (Section 5). Each criterion, with its meaning and rating source, is defined once in a shared catalogue, and criteria such as credit risk or compliance status serve both views. Each view has its own configuration of criteria, dimensions, weights, classes, escalation rules and actions. Internally every view computes 0 = favourable … 100 = unfavourable; the performance view is only *displayed* as higher-is-better. Expert labels, weight versions, runs and reviews are kept per view, so calibrating one view never changes the other. Everything in this section applies to each view; the experiments in Section 7 use the risk view. In both templates calibration is disabled by default and enabled per run; the performance view has not been calibrated in any experiment.

6.7 Actions, data providers and history

Action framework. Recommendations come from three levels, all configured per view: per class with a review cycle; per elevated dimension score; and per factor and level with a typical owner, where the most

specific level reached applies. For example, availability of alternatives rated 4 leads to a market scan, rated 5 to qualifying an alternative source. The actions of a portfolio form an exportable action plan; tracking to closure is not implemented. **Data providers.** A slim interface fills indicators missing from the supplier data and records the origin and vintage of each value. Hand-entered values are never overwritten, and a failing provider is reported without stopping the assessment. Bundled are a political stability provider reading a user-downloaded table (a fictional table, marked as such, serves the sample data), a generic provider for tabular exports from ERP, ESG-platform or credit-agency systems, and documented placeholders for API integrations. Providers deliver indicators only; rating stays with the rating sources, so the method does not change when a source does. **History.** Assessments can be recorded as timestamped snapshots per view, explicitly or automatically after an approved review, with scores, classes, dimension scores, weights and approver, so that suppliers and material groups can be followed over time. Scores can be recomputed at any time; snapshots are therefore not taken on every recalculation, which would only add noise. **Spreadsheet surface.** A browser interface presents all of this as a sheet, with dashboards per material group, the action plan and the history; all numbers come from the package.

6.8 Engineering properties

The full suite has 199 automated tests: 198 pass, and one smoke test against the real language-model API is skipped without an API key. They cover scoring and weights in both views, calibration (including recovery of known weights from noise-free labels), the weight store and review flow, rating sources, providers, actions, reporting, history, the interface, the consistency of the benchmark data, the reproducibility of the synthetic datasets, the optional review agent (with mocked language-model responses) and Explain & correct (contributions adding up to the score for all twelve supplier–view pairs of the sample data, redistribution for missing data, escalation effects, trace ids, correction kinds, mandatory reasons, rejection of invalid corrections, a preview that changes nothing, recorded corrections, disputes that change nothing, and the outdated marker, also in the interface). Continuous integration runs linting and tests on two Python versions; the checkpointer restores only allowlisted state types. The code is MIT-licensed; all data is fictional or synthetic and labelled as such.

6.9 Supplier Review Agent: optional, LLM-supported interpretation

One node, `ai_review`, uses a language model. It is the only one, and it stands outside the evaluation: it interprets results that the deterministic workflow has already produced (Table 3). *Trigger*: only an explicit user request (a "Generate AI Review" button, or a command-line option with confirmation) starts it, as a one-shot follow-up step on an existing run; a pending human review stays pending. *Input*: one supplier's deterministic scoring trace from both views (Section 6.10) — per criterion inputs and source, rating rule, rating and origin, nominal and effective weight and contribution; dimension scores, overall score and class, the class before escalation, non-additive rules with their effect, data completeness, missing criteria and the rule-based actions, each with a citable id. Supplier name, country, spend, the raw indicator record and expert labels are not sent. The payload is shown before sending, and the node checks that it still matches the run's results. *Output*: a structured answer without numeric fields — a management summary, main drivers (criterion and view), differences between the views, prioritised action proposals and open questions; the summary, every driver, view difference and action must cite the trace ids it relies on. The node writes only this review into the run and cannot change scores, classes or weights; scores shown with the review are copied from the deterministic results. *Safeguards*: the whole answer is rejected if it cites a criterion not in the input, does not compare the views, leaves missing data without an open question, or has a summary, driver, view difference or action that cites no trace entry or an unknown id; actions are labelled as proposals for human review; each attempt is recorded in the audit trail with a hash of the payload; failures are reported and leave the run usable. *Limits of validation*: the free text can still contain numbers, and a misquoted score or weight is not detected; validation checks that cited ids exist, not that a cited entry supports the sentence; whether statements are supported by the payload, whether explanations and priorities are sensible, and whether every missing criterion is addressed are not checked. The prompt and the withheld supplier name reduce, but do not rule out, invented facts. *Provider*: Anthropic's Claude, configured through environment variables; without an API key the feature is unavailable and everything else works unchanged. The component is experimental and unevaluated (Section 10).

6.10 Explain & correct: scoring trace, explanation and controlled correction

A category manager who receives a score asks *why this result?* and *what if an input is wrong?* The Explain & correct workflow answers both without giving up determinism, in four parts, each a section of an interface tab.

1. Scoring trace (deterministic). For each supplier and view, a trace is derived from the engine's own results; it needs no API key. Per criterion it lists the inputs with source and data date (a missing input is flagged "missing: no value and no source", a value without provenance "entered by hand (no source or data date recorded)"), the rating rule as text generated from the same band constants the rating functions use, the rating and its origin (expert assessment, expert override of a derived rating, derived from in-

Table 3. Deterministic evaluation and LLM-supported interpretation.

	Deterministic evaluation	**LLM-supported interpretation**
Nodes	Data collection, scoring, calibration, recommendation, human review, supervisor	Supplier Review Agent
Runs	In every run	Only on explicit request
Produces	Ratings, scores, classes, weights, rule-based actions	Summary, drivers, view differences, action proposals, open questions
Changes results	Yes, recomputably	No; scores shown are copied, numbers quoted in its text are not checked
Reproducible	Same input, same output	No; output can vary
Tested	Unit and flow tests	Mocked responses and real smoke tests on synthetic suppliers; quality not evaluated

put data by rule, or missing), the nominal weight θ_f, the effective weight e_f and the contribution c_f. With W_d the weight of dimension d and r_f the 1–5 rating,

$$e_f = \frac{\theta_f}{\sum_{g \in d,\ \text{rated}} \theta_g} \cdot \frac{W_d}{\sum_{d' \ \text{with data}} W_{d'}},$$
$$c_f = e_f \cdot \frac{r_f - 1}{4} \cdot 100$$

in internal points (0 = favourable … 100 = unfavourable). The contributions add up exactly to the internal score, which is tested for all six sample suppliers in both views; missing data is redistributed as in scoring, so a missing criterion has no effective weight. Rules that do not add points are listed separately: escalation rules with their effect ("raised the class from X to at least Y", "triggered, no effect", "not evaluable"), the data-completeness check and a class set by a reviewer. Every entry has a citable id, such as `risk:credit_risk`, `risk:rule:compliance_status` or `performance:dim:delivery`.

2. AI explanation (optional). The Supplier Review Agent receives this trace as its payload and must cite trace ids for its summary, drivers, view differences and actions (Section 6.9). It explains; it computes nothing.

3. Controlled correction. A *data correction* changes an input value, or the expert rating of an expert-assessed criterion; an *expert override* replaces a rating derived by rule with an expert 1–5 rating. The two are kept apart, and a reason is mandatory. Weights, thresholds and rules cannot be changed here; they stay with calibration and approval. A preview rescores the supplier deterministically in both views and shows before and after: score, class, class from score, completeness, review flag, changed criteria with rating, origin and contribution, and rules now or no longer triggered. Nothing is applied until confirmed. A confirmed correction is appended to a history with old and new values, kind, reason, time, the name as typed (not verified), model and weight version, and before/after per view. An existing AI explanation is then marked *outdated*, because its input hash differs, and is not regenerated automatically. The same path applies to the spreadsheet-like input table: first data entry is direct, but a changed cell of a supplier that already has a result is held back as a pending correction and applied only after a reason, the preview and confirmation.

4. Dispute. An objection (criterion, reason, status *open*) can be recorded without a corrected value; it changes no score and trains no weights.

The workflow separates explaining a result, correcting its inputs and changing the method, which a single editable cell would mix; only the first two happen here, and both leave a record.

7 Demonstration on synthetic data

All numbers in Sections 7.1–7.8 are produced by deterministic scripts in the repository, run on commit `6bb05b9` and re-checked on `c9a500f` (Section 7.9 reports one interface run on `b8a08b5`); the tables behind every figure are in `docs/paper/results/`. **Scope of the evidence:** the data is synthetic throughout. Every "expert label" in E1–E5 is generated by a constructed expert policy; no real category manager provided labels, and no result has been validated in the field. The experiments show how the procedure behaves under known conditions, not how well it works in practice.

7.1 Design

Generator. Synthetic suppliers are drawn with random ratings and indicators. Each receives an "expert label": its score under a **hidden weighting**, plus Gaussian noise ($\sigma = 4$), rounded to steps of five as category managers tend to state judgements, and clipped to $[0, 100]$. The hidden weighting differs from the template: the simulated experts care more about business impact (22 % instead of 14 %) and single sourcing (16 % instead of 8.75 %) and less about the ESG rating (3 % instead of 12 %). Noise and rounding imply an **irreducible error of 18.1 points²** (noise floor) that no model can beat out of sample. Because the generating weights are known, calibration can be judged on whether it reduces error out of sample, whether it moves the weights towards the truth, and how both depend on λ and n. **Data.** A reference portfolio of 36 suppliers (32 labelled), shipped with the repository; training draws of 8–192 labels (30 seeds per size, plus 100 seeds at $n = 32$); and an independent test set of 2,000 suppliers, scored also against noise-free true

Table 4. E1: repeated 5-fold cross-validation on the reference portfolio (32 labels, 20 repeats). MSE in points^2; noise floor 18.1; default weights: 37.8 out of fold and 84 % out-of-fold class agreement. ±: standard deviation across repeats. *Synthetic data.*

λ	CV MSE	in-sample	class agr.	L1 to hidden
0	22.9 ± 4.2	10.0	81 %	0.155
0.02	20.8 ± 2.7	10.7	83 %	0.142
0.05	$\mathbf{20.6 \pm 2.0}$	12.4	86 %	0.163
0.2	24.4 ± 1.2	18.9	84 %	0.241
1	32.3 ± 0.5	29.9	84 %	0.309
5	36.4 ± 0.1	35.8	84 %	0.337

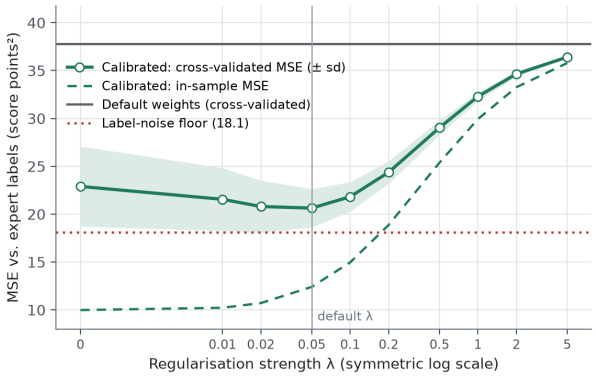


Figure 3. Cross-validated and in-sample MSE of calibrated weights versus λ on the reference portfolio. Horizontal lines: cross-validated error of the default weights and the irreducible label noise. The gap between the solid and dashed lines is the in-sample optimism. *Synthetic data.*

scores to isolate estimation error. **Experiments.** E1: repeated 5-fold cross-validation [23, 18] on the reference portfolio over λ; E2: sample size $\times$ λ on the test set; E3: weight stability over 100 seeds; E4: misspecified expert policies; E5: the acceptance threshold.

7.2 Reference run

On the reference portfolio with $\lambda = 0.05$, the error between score and label falls from 37.8 to 13.3 points^2, class agreement rises from 84 % to 88 %, the L1 distance to the hidden weighting falls from 0.345 to 0.138, and re-scoring with the calibrated weights (applied only after approval under the current risk template) changes the class of 2 of 36 suppliers. The acceptance rule accepts the calibration on an out-of-fold error of 37.8 $\rightarrow$ 22.8 ($z = 2.4$). The in-sample 13.3 lies *below* the noise floor: ten weights fitted to 32 noisy labels partly fit the noise. E1 quantifies by how much.

7.3 Out-of-sample error and the choice of λ (E1)

Calibration generalises (Table 4, Figure 3): at $\lambda = 0.05$ the cross-validated error falls from 37.8 to 20.6, a 45 % reduction within 2.5 points^2 of the noise floor, while the in-sample error overstates the gain by about

Table 5. E2: test error against true scores (mean over 30 seeds; default weights 29.7) and, in brackets, the share of seeds in which calibration beats the defaults. Fixed λ, the scaled default $0.05 \times 32/n$, and λ chosen by cross-validation with the minimum-error rule. *Synthetic data.*

n	$\lambda = 0$	$\lambda = 0.05$	**scaled**	CV min
8	37.7 (50 %)	16.7 (90 %)	**18.4 (100 %)**	24.5 (90 %)
16	16.2 (90 %)	9.9 (100 %)	**11.5 (100 %)**	12.8 (100 %)
32	6.1 (100 %)	5.9 (100 %)	**5.9 (100 %)**	5.6 (100 %)
64	2.8 (100 %)	4.2 (100 %)	**3.0 (100 %)**	3.0 (100 %)
192	0.8 (100 %)	2.5 (100 %)	**0.8 (100 %)**	0.8 (100 %)

8 points^2. The default lies at the bottom of the curve, but it was set before this experiment on data from the same generator, so this is consistency, not proof. Regularisation buys stability more than accuracy: from $\lambda = 0$ to 0.05 the mean improves modestly and its spread halves; strong regularisation gives up most of the gain. Error and class agreement can diverge: without regularisation, out-of-fold class agreement *falls* below the default (81 % vs. 84 %) although the error improves — which is why both are reported.

7.4 How many labels, and how strongly to regularise (E2)

Against the noise-free true scores, the default weights have a structural error of 29.7 points^2 (Table 5, Figure 4).

Small samples need regularisation. With eight labels and no regularisation, calibration is *worse* than the defaults on average (37.7 vs. 29.7) and wins in only half of the seeds; with $\lambda = 0.05$ the same labels nearly halve the error and win in 90 %. At 32 labels the default removes about 80 % of the structural error (29.7 $\rightarrow$ 5.9). **A fixed λ becomes a bias at large n**: beyond about 48 labels the unregularised fit is better (0.8 vs. 2.5 at $n = 192$), because the error term is a mean over suppliers while the penalty is not scaled. **Scaling λ with $1/n$** — a prior of constant strength, anchored at $\lambda = 0.05$ for 32 labels — removes this bias (0.76 vs. 0.78 at $n = 192$), beats the defaults in every seed from eight labels upwards, and under the default acceptance rule caused no harmful acceptance at any size. Choosing λ by cross-validation was tested and not adopted: the one-standard-error rule [9] over-regularises in the middle range (10.6 vs. 5.9 at 32 labels), the minimum-error rule is too noisy with few labels (24.5 vs. 18.4 at eight), and both cost an order of magnitude more computation. The scaling constant was itself set on the same synthetic data; it is a reasoned default.

7.5 Stability of the recovered weights (E3)

Across 100 portfolios of 32 labels ($\lambda = 0.05$), the mean L1 distance to the hidden weighting is 0.170, against 0.345 for the defaults (Figure 5). Large deviations are recovered reliably: business impact (14 % $\rightarrow$ 22 %, calibrated mean 19.5 %), sourcing

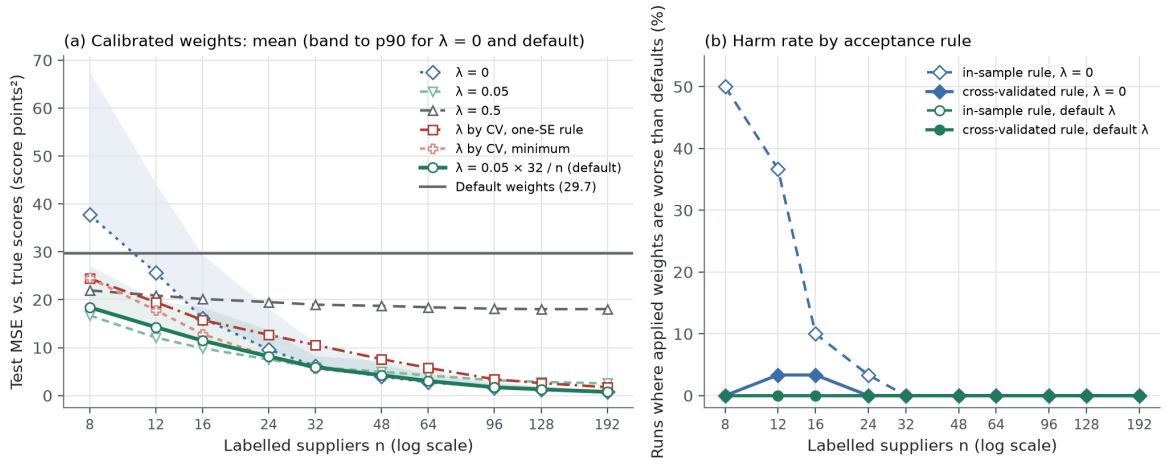


Figure 4. (a) Test error of calibrated weights against noise-free true scores, by number of labels, for fixed λ, the scaled default and λ chosen by cross-validation (mean over 30 seeds; band to the 90th percentile). (b) Share of runs in which the weights actually applied are worse than the defaults, under the in-sample rule and the default cross-validated rule (5 % and $z \geq 2$). *Synthetic data.*

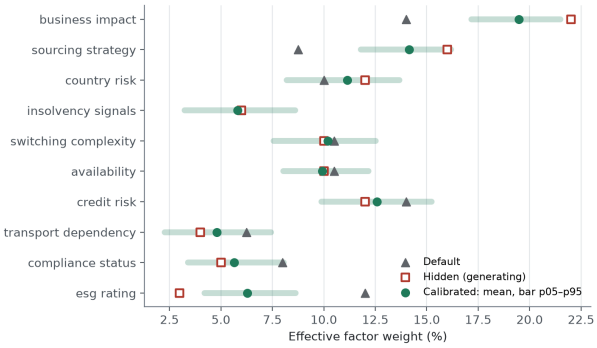


Figure 5. Effective factor weights: configured default, hidden weighting, and calibrated weights (mean and 5th–95th percentile over 100 portfolios of 32 labels, $\lambda = 0.05$). *Synthetic data.*

strategy (8.75 % → 16 %, 14.1 %) and ESG rating (12 % → 3 %, 6.3 %) move in the right direction in all 100 seeds. Small deviations are not: for two factors whose default and hidden weights differ by half a percentage point, the direction is right in only 67 and 54 seeds, close to chance. Estimates are shrunk towards the default as intended; large shifts are recovered at roughly two-thirds of their size.

7.6 Acceptance rules: in-sample versus cross-validated

The demonstrator originally accepted calibrated weights if they reduced the *in-sample* error by at least 5 %. In E1–E3 this rule accepted practically every calibration, including the eight-label, unregularised runs in which the weights were worse than the defaults in half of the seeds. **An in-sample rule is not a safeguard against overfitting**; it only protects against calibrations that change nothing. It was replaced by an out-of-sample test: refit on $k - 1$ folds, evaluate on the held-out fold, repeat with reshuffled folds (5×3, deterministic), and require the pooled out-of-fold error to improve by at least 5 %. Table 6 reports this 5 % rule (commit `ddcb977`); Section 7.8 adds the second condition that is now the default.

Table 6. In-sample versus 5 % cross-validated acceptance on the E2 runs (30 seeds; error against true scores; defaults 29.7). Harm: share of runs in which the weights in use are worse than the defaults. *Synthetic data.*

n	λ	accepted in-s. / CV	error in use in-s. / CV	harm in-s. / CV
8	0	100 / 23 %	37.7 / 30.8	50 / 7 %
8	0.05	100 / 57 %	16.7 / 20.8	10 / 3 %
12	0	100 / 47 %	25.7 / 27.7	37 / 20 %
12	0.05	100 / 70 %	12.1 / 16.3	7 / 3 %
16	0.05	100 / 87 %	9.9 / 11.8	0 / 0 %
≥ 24	0.05	100 / 100 %	identical	0 / 0 %

The cross-validated rule sharply reduces harm where harm occurs (50 % to 7 % at eight unregularised labels). It is conservative: with few labels it also rejects helpful calibrations, so the error of the weights in use is higher than under the in-sample rule (20.8 vs. 16.7 at $n = 8$), though still well below the defaults. It trades expected gain for a lower risk of making things worse — the appropriate trade-off for weights that drive supplier decisions. From about 24 labels both rules agree. On the shipped data, the rule changes one visible outcome: a calibration restricted to the ten labelled "plastics" suppliers looked excellent in-sample (36.8 → 10.4) but was worse out of sample (36.8 → 41.7) and is now rejected. Group-specific calibration needs enough labels per group.

7.7 When experts do not judge like the model (E4)

E1–E3 generate labels from the linear model family that calibration fits: the best case. E4 replaces that policy with alternatives a real panel might fol-

low: more or less noise, coarser rounding, a uniform level shift (10 points stricter), an interaction between business impact and single sourcing, a knock-out floor (a severe compliance or insolvency rating lifts the judgement to at least 70), a non-compensatory "worst factor dominates" policy, and two experts with different weightings. For each policy, an **approximation floor** — the error of the best constrained linear weighting on 2,000 noise-free judgements — measures how far it lies outside what weights can express. Calibration then runs with production settings on 16, 32 and 64 labels, 30 seeds each (Table 7, Figure 6).

Noise and rounding cost labels, not validity: with linear experts, calibration stays effective under coarse rounding and doubled noise, but needs more labels. **Non-linear policies leave a floor no weighting removes**: for the interaction, knock-out and worst-factor policies, calibration closes much of the gap above the floor with 32–64 labels, but class agreement barely improves. Weights cannot learn a policy that is not additive, and a large remaining error should be reported as exactly that signal. **Knock-outs belong in escalation rules, not in weights**: with sixteen labels and a fixed λ, calibration is worse than the defaults on average, and even the 5 % rule lets 10–13 % of runs end with harmful weights (0–3 % under the current default, mostly because such calibrations are no longer accepted). The demonstrator's escalation rules are the right mechanism; knock-outs should be *elicited* from category managers and encoded as rules. **A level offset cannot be expressed by weights**, which sum to one. Calibration then distorts the weights to imitate the offset: weight moves towards factors that are usually rated high (the shift correlates at 0.89 with the factor's mean rating), and insolvency signals and compliance status — the rare, severe drivers a risk model should protect most — are pushed to the minimum. The demonstrator now diagnoses this: it flags a mean difference of at least 5 points that is also two standard errors from zero and withholds such calibrations for review. The flag caught every level-shift and worst-factor run, 30–37 % of knock-out runs, at most 7 % of purely linear runs at sixteen labels and none from 32. **Disagreeing experts are the hardest case**: their disagreement acts as structured label noise, and the 5 % rule accepted harmful weights in 43 % of runs at sixteen labels — the motivation for E5.

Fit diagnostics. Since knock-outs and non-additive judgement cannot be fixed by weights, the report says when they are likely, without blocking approval. A *high residual* (cross-validated error at least 40 % of the label variance) detects 63–100 % of non-linear runs and 0–7 % of linear ones. A *knock-out candidate* (at least three suppliers with a severe rating judged at least 5 points and two standard errors riskier than the model predicts, after removing the offset) found the knock-out policy in 13/53/97 % of runs at $n = 16/32/64$, always on the right factor family, with at most 3 % false hints from linear experts (7 % with very noisy labels). A factor *pushed to the minimum weight* is reported only as "check this factor": it is unspecific. None of the hints detects expert disagreement reliably. E4 thus maps where calibration works (noisy but broadly additive judgement), where it is partial (non-additive policies) and where it fails (knock-outs, level offsets, disagreement) — but it cannot say which policy real category managers follow.

7.8 Requiring a clear improvement (E5)

A fixed 5 % threshold cannot tell a real improvement from a lucky one when the estimate itself is noisy. The acceptance rule therefore has a second condition. Cross-validation yields an out-of-fold error for every labelled supplier under the active weights and under refitted weights; from the per-supplier differences the rule computes the mean improvement and its standard error, whose ratio z must reach a threshold. This standard error is optimistic, because out-of-fold errors are not independent [20, 2]; z is a screening statistic, not a test with a nominal error rate, and the threshold was chosen **empirically**. E5 covers 20 scenarios (five expert policies × 8/16/32/64 labels), seven thresholds and 100 seeds each, with the scaled λ (Table 8).

The selection criterion was the *smallest threshold for which no scenario exceeds 5 % harmful acceptances*: $z \geq 2$. Against the 5 % rule alone, it gives up about nine percentage points of the default error in expected improvement (0.78 → 0.87) for a worst-case harm rate that falls from 30 % to 4 %. Re-run with the new default, harm falls to zero for disagreeing experts (from 43/23 % at 16/32 labels), to 0/3 % for knock-outs, and to 0 % at every size in E2 with the scaled λ; the reference portfolio is still accepted ($z = 2.4$). **Calibration now has to be earned**: with linear experts, 17 % of calibrations are applied at 8 labels, 50 % at 16 and 87 % at 32; with disagreeing experts, none at 16 or 32. When it withholds a calibration, the rule keeps the defaults and states why. For practice this means roughly **30 or more consistent labels per calibration scope**, and measuring expert agreement before calibrating a scope with several labelling experts. A rule that sometimes forgoes an improvement is the right failure mode for weights that drive supplier decisions; a rule that sometimes makes them worse is not. The threshold was selected and evaluated on the same synthetic scenarios and should be treated as a reasoned default.

7.9 Explain & correct on one synthetic supplier

This case was run end to end in the interface on 12 October 2026 (commit `b8a08b5`) on the fictional supplier SUP-004 "Delta Fibre Media". All its inputs are entered by hand without source or data date, and the trace says so (Figure 7).

Before. The risk score is 41.75, in the band *medium*; the final class is *high* because the escalation rule `risk:rule:compliance_status` fired (compliance status *critical_findings*, rating 5). The largest con-

Table 7. E4: calibration under misspecified expert policies. Errors in points^2 against the policy's noise-free judgement, "calibrated" at $n = 32$. Excess removed: share of the default weights' error above the linear floor that calibration removes, at $n = 16/32/64$. Harm: share of runs with weights in use worse than the defaults, at $n = 16/32$. Original: fixed $\lambda = 0.05$ and the 5 % rule; current: scaled λ and the rule with $z \geq 2$. *Synthetic data.*

Policy	floor	default	calibrated	excess removed		harm		class agreement
				original	current	original	current	default → cal.
linear (reference)	0	29.7	5.7	69 / 81 / 87 %	62 / 81 / 91 %	0 / 0 %	0 / 0 %	79 → 91 %
linear, $\sigma = 8$	0	29.7	11.7	30 / 61 / 75 %	39 / 61 / 76 %	7 / 0 %	0 / 0 %	79 → 87 %
linear, steps of 10	0	29.7	7.0	63 / 76 / 84 %	58 / 76 / 88 %	0 / 0 %	0 / 0 %	79 → 90 %
level shift +10	62.3	157.4	77.6	74 / 84 / 90 %	77 / 84 / 91 %	0 / 0 %	0 / 0 %	44 → 60 %
interaction	71.9	103.1	88.0	9 / 48 / 68 %	24 / 48 / 66 %	10 / 0 %	0 / 0 %	63 → 65 %
knock-out floor	98.3	127.0	118.8	−24 / 28 / 62 %	8 / 28 / 58 %	10 / 13 %	0 / 3 %	66 → 69 %
worst factor dominates	284.3	418.0	328.1	57 / 67 / 82 %	69 / 67 / 78 %	0 / 0 %	0 / 0 %	12 → 27 %
two experts disagree	0	5.3	5.3	−84 / 1 / 42 %	−22 / 1 / 29 %	43 / 23 %	0 / 0 %	90 → 91 %

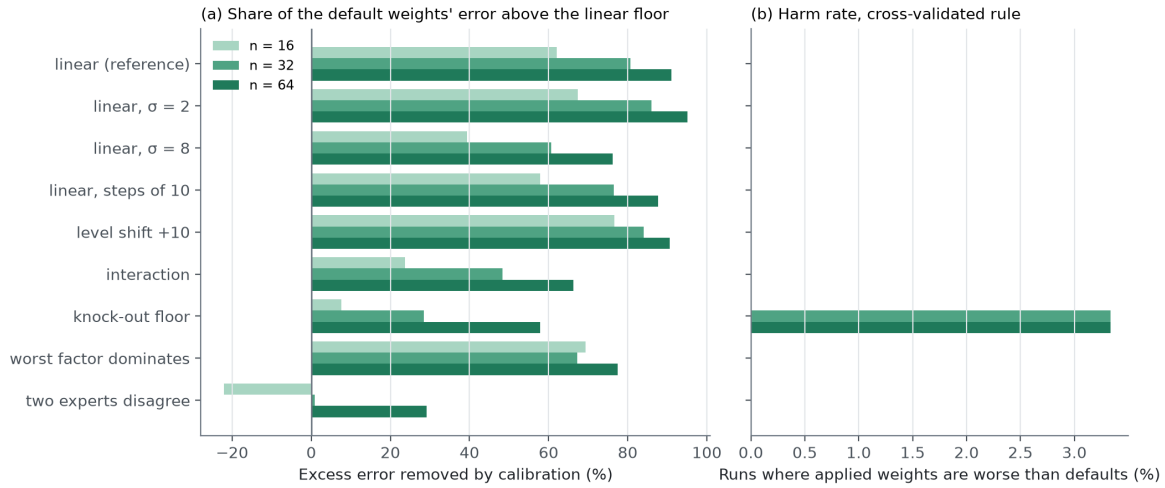


Figure 6. (a) Share of the default weights' excess error above the linear floor that calibration removes, by expert policy and number of labels. (b) Share of runs in which the weights in use are worse than the defaults under the default rule (5 % and $z \geq 2$); almost all bars are zero. *Synthetic data.*

Table 8. E5: effect of the uncertainty threshold, averaged over 20 scenarios (100 seeds each). Missed gain: error of the weights in use minus the better of default and calibrated weights, as a share of the default error. Rel. error: error of the weights in use divided by the default error. *Synthetic data.*

$z \geq$	harm mean / worst	accepted	missed gain	rel. error
0 (5 %)	7.4 / 30 %	61 %	10.1 %	0.78
1	4.2 / 17 %	46 %	12.4 %	0.80
1.5	2.2 / 8 %	34 %	15.2 %	0.83
2 (default)	**1.1 / 4 %**	24 %	18.7 %	0.87
2.5	0.4 / 3 %	15 %	22.2 %	0.90
3	0.1 / 1 %	10 %	25.1 %	0.93

tributions are the ESG rating with 12.0 points (ESG score 22, rating 5), compliance status 8.0, change complexity 5.25 (an expert override) and sourcing strategy 4.38; all contributions sum to 41.75. The performance score is 38.75 (internal 61.25), class C, driven by on-time delivery and delivered quality with 22.5 internal points each; the compliance escalation fired there too but had no effect, as the class was already C.

AI explanation. A real explanation (model `claude-opus-5-5`) passed validation. In a manual spot check, the author found all 64 numbers in its text in the trace. It explained the escalation separately from the score, named the hand-entered inputs without source, and asked open questions for the three missing inputs (time to exit, political stability, intercontinental transport). Its action "suspend new business until resolved" is one of the rule-based actions in the trace.

Correction and preview. A data correction set compliance status from *critical_findings* to *minor_findings*, with the reason "Re-audit on 30 Sep 2026 closed the critical finding; two minor findings remain (synthetic example)", entered by "Category manager (example)" (not verified). Table 9 shows the preview. After confirmation, the correction was recorded with these values, the model version (`generic-supplier-risk-template@0.1.0` and the performance template) and the weight version (`global/default`); the AI explanation was marked outdated, and no new language-model call was made.

What the case shows, and what it does not.

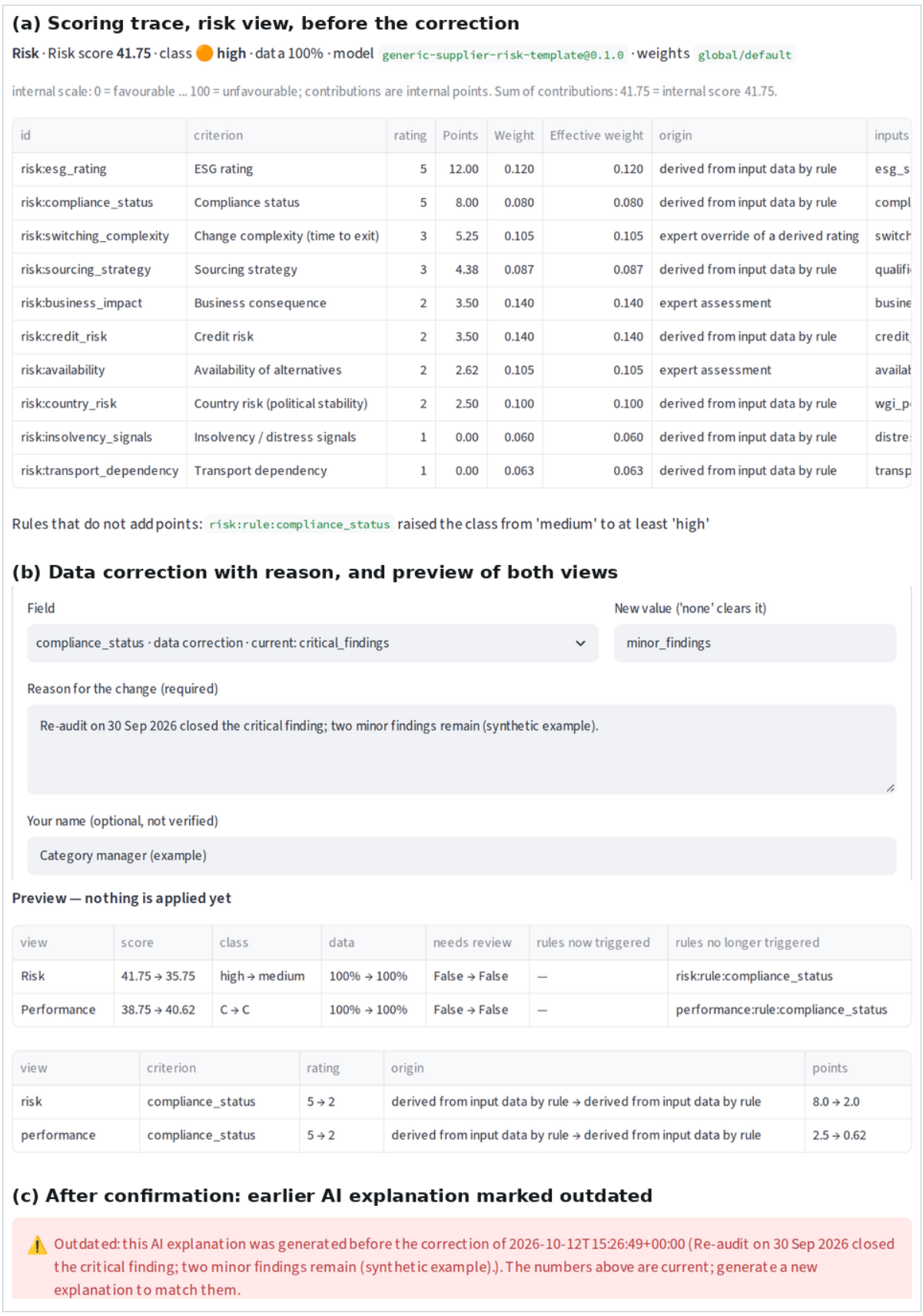


Figure 7. Explain & correct for the fictional supplier SUP-004 (interface screenshots, synthetic data). (a) Scoring trace of the risk view before the correction: rating, contribution in points, nominal and effective weight and rating origin per criterion; the points add up to the internal score of 41.75 (inputs and rule columns are cut off on the right). The escalation is listed separately as a rule that adds no points. (b) Data correction with a mandatory reason, and its preview: both views re-scored, nothing applied yet. (c) After confirmation, the earlier AI explanation is marked outdated and not regenerated.

Table 9. Preview of the correction for SUP-004 (synthetic). Performance scores as displayed (higher is better); contributions in internal points.

	Risk view	**Performance view**
Score	41.75 → 35.75	38.75 → 40.62
Class	high → medium	C → C
Compliance status: rating	5 → 2	5 → 2
Compliance status: contribution	8.0 → 2.0	2.5 → 0.62
Compliance escalation	no longer triggered	no longer triggered

The parts work together as designed on one synthetic supplier: the trace accounts for the score exactly and keeps the class floor apart from it, the correction removes the escalation and lowers the risk class, the effect is visible before anything is applied, and the record and the outdated marker follow confirmation. Together with the tests, this is **technical verification**: the arithmetic adds up, the schema and citation checks reject non-conforming answers, and the real API path works. It is not **domain validation**, which has not been done: no practitioner has used the workflow, no real supplier data was involved, and neither explanation quality, nor correction accuracy, nor the effect on false alarms or decisions was measured. Whether the re-audit justified the correction is the judgement the workflow leaves to a person; the system only makes it explicit and recorded.

7.10 What the demonstration shows, and what it does not

Under controlled conditions, the procedure generalises close to the noise floor, recovers the direction and roughly the size of substantial weight differences, needs regularisation when labels are few, and shows the user in-sample figures that are optimistic by a quantifiable margin. It does not show how real expert judgement behaves: E1–E3 use best-case labels, and E4 cannot say which policy describes real category managers or whether it changes over time. Policy-capturing research suggests that linear models often describe holistic judgements well [15]; whether this holds for supplier-risk judgements is an empirical question. The demonstration also says nothing about **predictive validity**.

8 Comparison and discussion

8.1 What the benchmark and the demonstrator show together

The benchmark locates a gap rather than a winner. Simple, manually maintained spreadsheet scorecards keep the method in the procurement team's hands but typically do not hold it to account; platforms bring data, scale and operational maturity but keep the method partly or largely outside the customer's view; open-source agent projects offer architecture, with little emphasis on a scoring method. In the author's qualitative assessment, no approach is strong both on *data coverage* and on *ownership of the method*, and none of the established ones documents publicly an evaluation of risk and performance on one basis. The demonstrator shows that the method layer can be built lean and can carry both views: deterministic and tested, orchestrated by deterministic, specialised agents over a shared state (no language model in scoring, calibration, recommendations or routing), governed by a real pause and versioned approvals, explainable and correctable per result through a trace whose contributions add up to the score and reasoned, previewed, recorded corrections, able to learn under a cross-validated acceptance test, with risk management, performance evaluation and an action framework from one catalogue — in a small, MIT-licensed code base. What neither part shows is that real category managers judge in a way a linear model can capture, that calibrated scores predict disruptions, that the approach scales operationally to an enterprise portfolio, or that the trace, its explanations and the corrections are useful in practice, reduce false alarms or improve decisions.

8.2 The hybrid reading: signals from outside, method in-house

The most useful reading of Figure 1 is not "which approach is best" but "which layer each approach should provide" (Table 10). In this reading the demonstrator does not compete with a platform; it *consumes* one. The integration points exist: rating sources turn any indicator, including a platform score, an alert count or a country index, into a 1–5 rating; a pluggable data source delivers indicators; and data providers fill missing indicators with recorded origin — the political stability provider is a small instance of a signal layer built on open data. Three consequences follow. *Platform scores become inputs, not verdicts*: procurement decides, and calibrates, how much they count per material group, and the class remains recomputable. *Double counting must be managed*: a platform score that already contains country risk overlaps with a separate country factor; either one is dropped or calibration is left to down-weight it, which works only with enough labels. *Licence terms matter*: whether platform scores may be stored, versioned and recomputed in-house depends on the contract, which this paper did not examine.

8.3 When which approach fits

Because the profiles are not ranked, the choice depends on the situation. A *spreadsheet with discipline* — written definitions, versioned weights, a fixed review date — is adequate for a small portfolio with stable risk drivers and no documentation duty. A *commercial platform* is hard to replace where tier-N exposure, real-time monitoring or regulatory evidence dominate; building that coverage in-house is neither

Table 10. A layered reading of supplier risk evaluation.

Layer	Question	Strong here / suggested provider
Signal	What is happening to this supplier, its country, its tier-N network, now?	Platforms (signals, coverage); subscription or open data (governance indicators, sanction lists, insolvency registers)
Method	How do signals and internal knowledge become a class, with which weights, and why?	Lean Agentic (transparency, calibration, group weights); in-house, owned by procurement
Decision	What do we do about it, and who decides?	Shared; category managers and a named decision owner, see Section 9

lean nor realistic. A *Lean Agentic method layer* fits organisations that have outgrown a simple spreadsheet scorecard, want to own and justify their method, need material-group weighting and want their category managers' judgement to shape the model — also alongside a platform. For a mid-sized procurement organisation, the most likely configuration is the combination.

8.4 The cost of being lean

"Lean" and "open source" do not mean free. Cost shifts from licences to ownership (someone maintains configuration, code and data connections), skills (a configuration file to change the method, Python to extend the system) and responsibility (no vendor, no service levels, no security hardening; operational maturity is rated 2). The approach is lean in what it *builds*, not necessarily in what it *requires* to run.

8.5 What the "agentic" part contributes

Scoring, calibration, recommendations and routing use no language model at all; the one language-model component, the optional review agent, interprets results after the fact (Section 6.9). For the current scope, the agent architecture is not strictly necessary; the same logic could run as a pipeline. It pays off as the system grows: the supervisor routes bounded loops explicitly, foreseeable additions (an external-signal agent, a document-extraction agent, an explanation agent) become new nodes rather than rewrites — the review agent is a first, experimental instance of the last, added as one node without changing the others — and checkpointing provides the pause-and-resume that human review needs. The value of "agentic" here is **orchestration and extensibility, not autonomy**: authority over scores and decisions stays with code that can be recomputed and with people who can be named.

9 Governance and decision ownership

A risk score does not decide anything; it informs a decision made by someone accountable for it. Table 11 proposes how roles could be assigned; the demonstrator does not enforce them.

Two principles follow. **The method owner sets the floor**: escalation rules express non-negotiable positions and are not subject to calibration, which optimises agreement with average judgement, while floors exist precisely for the cases averages hide. **Calibration proposes, people decide**: for weights that drive supplier decisions, the recommended setting is to propose rather than auto-apply (`auto_apply: false`), the default of both bundled templates (risk and performance), with the approver seeing the number of labels, the cross-validated and in-sample error, and old and new weights. **Separation of duties**: the people whose labels drive a calibration should not be the only ones to approve it, or a team could steer the weights by the way it labels; the demonstrator records approvers but has no role model to enforce this.

A calibration cycle in the supplier review. (1) Category managers record a holistic score for the suppliers they know well, *before* seeing the model score, to avoid anchoring. (2) The portfolio is scored and calibrated per scope, accepted or rejected on cross-validated error. (3) The approver reviews proposed weights and flagged suppliers; approved weights become a new version, and a snapshot records the result. (4) Weight changes and the class changes they cause are communicated, since a weight change that moves suppliers between classes is itself a decision. (5) In the next cycle the regularisation still anchors on the method owner's defaults, so repeated calibration does not drift.

Risks to governance. *Label gaming* — mitigated by blind labelling, separation of duties, reporting how much each weight moved, and the minimum weight, which prevents criteria from being calibrated away. *Label drift* — versioned weights and the assessment history make it visible; they do not prevent it. *Automation bias* — the completeness flag and the stated number of labels are partial counter- measures; the class, not the decimal score, should be the unit of communication. *Accountability gaps* — for due-diligence documentation the record must show who decided, on what basis, and what was done; the audit trail provides the basis, the decision owner remains responsible.

10 Limitations

Evidence. No field data supports the demonstration; all suppliers and labels are synthetic and labelled as such. E1–E5 are empirical only in the sense of controlled simulation: no real expert labels were collected and no result was validated in the field. E1–E3 use

Table 11. Proposed ownership of the elements of the method.

Element	Owner (proposed)	Changed how	Recorded where
Criteria, scales, default weights, thresholds, escalation rules	Method owner (e.g. procurement excellence)	Reviewed configuration change, new model version	Version control; model version on every result
Expert ratings per supplier	Category manager	During the supplier review	Supplier record; rating provenance per factor
Holistic labels for calibration	Category managers, blind to the model score	Periodic labelling round	Supplier record
Calibrated weights	Approver distinct from the label providers	Approval at the review pause	Versioned weight store with approver, metrics, timestamp
Manual class overrides	Decision owner for the supplier	Review decision with mandatory comment	Assessment; audit trail
Data corrections and expert overrides of derived ratings	Category manager	Explain & correct: mandatory reason, confirmed after preview	Correction history with old and new values, reason, name as typed (not verified), model and weight version
Disputes of an assessment	Anyone reviewing the result	Objection with reason; changes no score	Objection log, status *open*
Indicators from external sources	Data owner per source	Data refresh	Provider and data vintage per indicator; source data not versioned

labels from the fitted model family, the best case; E4 tests constructed alternatives and shows clear failure modes, but cannot establish which policy real experts follow. Agreement with expert judgement is not predictive validity. The literature was read at abstract level, and claims drawn from it are phrased accordingly.

Benchmark. The benchmark is the author's qualitative expert judgement, not an empirical measurement. The success factors and all ratings come from a single rater, the author, who also built one of the rated approaches and thus rates their own work; rationales, confidence marks and the v0.2 correction reduce this bias but do not remove it. The v0.3 review added a factor on which the proposed approach is strong (risk and performance integration), sharpened another towards agentic orchestration, and re-rated five factors in its favour after practitioner feedback. The v0.5 review sharpened factors 2 and 6 and wrote rating anchors after the author had built the Explain & correct workflow; the anchors fit the demonstrator's design and raised its ratings on both factors. These changes are documented, but they increase the need for independent re-rating. Where public documentation is silent, values are marked unknown or as documented minimums; the SaaS profile then averages only known values, and vendors may offer more. The platforms were rated on public information as of October 2026 — whether they offer performance evaluation or agentic workflows beyond their public documentation is not known — and the spreadsheet profile is a stereotype of a simple, manually maintained scorecard, not a survey; its ratings describe that profile, not what spreadsheets can do in general.

Method. The default acceptance rule ($z \geq 2$) is conservative — it withholds half or more of all calibrations with 16 or fewer labels — and its threshold was chosen and evaluated on the same scenarios, with a standard error that ignores the dependence between folds. The in-sample rule remains available and should not be used for real decisions. Knockouts, level offsets and non-additive policies cannot be learned by weights; offsets are withheld for review, the other hints were tuned on constructed policies, and disagreement between experts is not reliably detected. The λ scaling constant was set on synthetic data. The criteria, weights and thresholds are illustrative, including the bands for political stability and time to exit and the factor-level actions; none was calibrated or validated. The performance view uses the same engine but is covered by none of the experiments; calibration is disabled by default in both templates.

Explain & correct. The workflow is technically verified: exact additivity of contributions, the correction and history logic, schema and citation checks, and one real end-to-end run on a synthetic supplier. Domain validation has not been done: no practitioner has used it, no real supplier data was involved, and neither the quality of explanations, nor the accuracy of corrections, nor false alarms or decision quality were measured; no such benefit and no superiority over the commercial platforms is claimed. The trace is only as good as the data entered (in the sample data, all inputs lack a source); names in the history are typed in and not verified; corrections and objections are local files without access control; and objections have no resolution workflow.

Demonstrator. "Agentic" denotes an orchestrated workflow of deterministic agents with a rule-based supervisor; document extraction and an LLM-based supervisor are planned, not implemented. The optional *Supplier Review Agent* is experimental and unevaluated: it was tested with mocked model responses on synthetic suppliers, plus real smoke tests and manual spot checks of three synthetic suppliers (one complete, one with many data gaps, and SUP-004 with the trace payload, Section 7.9). These check plumbing, validation and the traceability of numbers in three answers, not quality in general; nothing here evaluates the quality, correctness or usefulness of the reviews, and no claim of domain validity follows. When triggered, it sends the structured results (without name, country or spend) to an external provider, Anthropic. Its validation checks references to criteria, views and trace ids and the handling of missing data, not every sentence and not whether a cited entry supports its sentence, so the text may still misread or overstate the results and should be read as a draft for a human reviewer. The demonstrator is not production-hardened: no roles or permissions, review pauses held in memory only, no live external signal feeds (the provider interface reads open data and exports, the bundled political stability table is fictional, and API integrations are placeholders), an assessment history in local files without access control, and no tracking of actions to closure.

11 Future work

Method. Validate the λ default on field data, possibly per material group. Turn diagnostics into proposals — a draft escalation rule for a knock-out candidate, a threshold shift for a level offset — reviewed like calibrated weights. Found the acceptance test better, with variance estimators that account for fold dependence [20] and validation on field data. Measure expert disagreement with repeated labels and calibrate only where agreement is adequate. Report bootstrap intervals for calibrated weights; Figure 5 shows that small shifts are not well supported. Allow ordinal labels, and partial pooling across material groups so that small groups borrow strength from the global fit.

Evidence. A field pilot in one or two material groups, with labels collected blind to the model score and a pre-registered comparison of cross-validated error with and without calibration. An independent re-rating of the benchmark by three to five practitioners against the published anchors, shown as bands in Figure 1, in particular of the v0.3 additions, the v0.4 consolidation and the v0.5 anchors for factors 2 and 6, which the author wrote after building the feature they reward; ideally including hands-on evaluation of the platforms. Predictive validation, linking scores to observed incidents over time — the only route to the question this paper leaves open.

System. *External data sources behind the provider interface*: the World Bank governance indicators retrieved online instead of from a downloaded table, an ESG rating platform for ESG scores and questionnaire status, an ERP system for on-time delivery and quality, and a credit agency for credit scores and insolvency signals, each recording the data vintage so that results stay recomputable. *An external-signal agent* on open data (sanction lists, insolvency announcements, event monitoring) as time-varying indicators. *Further views and a joint reading*: the view mechanism is not limited to two; a sustainability view or a view for strategic suppliers needs only a configuration on the shared catalogue, and two extensions are open — a joint portfolio reading of performance grade against risk class, and calibration of the performance view against its own labels with the same safeguards. *Evaluate the review agent with practitioners*: the Supplier Review Agent is implemented but unevaluated; category managers should rate its reviews on real or realistic cases for correctness, faithfulness to the results and usefulness, and the result should decide whether it stays. Document extraction and an LLM-based supervisor remain planned. *Evaluate Explain & correct with practitioners*: whether users can state why a supplier got its class, whether cited trace entries support the explanation's sentences (a check that could also be automated), how often corrections hold on later review, and whether false alarms are resolved faster. *Roles and persistence*: enforced separation of duties, verified identities in the correction history, a resolution workflow for objections, a persistent checkpointer, a central history store and tracking of actions to closure. *A platform integration pattern* that turns a platform score into a rating source and handles overlapping factors.

Benchmark maintenance. Repeat the public-information review of the platforms periodically with dated ratings, and add vendors whose methodology is publicly documented.

12 Conclusion

Supplier risk evaluation in mid-sized organisations often sits between a simple, manually maintained spreadsheet scorecard that the team owns but that typically does not hold the method to account, and a platform that brings data and scale but whose method the team does not fully own; open-source agent projects offer architecture but rarely a method. Across all three, the weights that turn criteria into a score are usually set once and never tested against the judgement of the people who know the suppliers.

This paper proposed fifteen success factors and used them to profile four approaches without ranking them. The comparison locates a gap rather than a winner: no approach combines data coverage with ownership of the method, and the established ones leave risk management and performance evaluation in separate systems. Lean Agentic AI addresses the method side

and closes the second gap: evaluate risk and performance as two views on one catalogue, start from existing scorecard practice, run the method as an agentic workflow of deterministic, specialised agents over a shared state with a rule-based supervisor and no language model in scoring, calibration, recommendations or routing (an optional, experimental review agent uses one only to interpret final results), keep every number deterministic, calibrate weights against expert labels under constraints with a cross-validated acceptance test, map results to concrete actions, pause for human approval wherever judgement is required, and make every result explainable and correctable — a deterministic trace whose contributions add up to the score, an optional language-model explanation that must cite it, and corrections with a reason that are recomputed deterministically, previewed before and after, confirmed by a person and recorded. On synthetic data, calibration generalises close to the noise floor and recovers substantial weight differences; with few labels it needs both regularisation and an out-of-sample acceptance test, and the validation found and fixed a rule that would otherwise have let harmful calibrations through.

What remains open is stated as plainly: the labels were synthetic, with no real expert labels and no field validation; the benchmark is the author's qualitative assessment, not a measurement; Explain & correct is verified technically, not validated with practitioners or real data; agreement with experts is not predictive validity; and the approach supplies external signals only in part. The most realistic deployment is therefore hybrid: signals from a platform or open data, and one method for risk and performance that is transparent, versioned, calibrated and owned by the procurement team that is accountable for it.

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Appendix A: Full comparison matrix

Table 12 lists every rating with its confidence and marks; the rationale and evidence for each cell are in `research/comparison-matrix.md`.

Data and reproducibility. All data is synthetic or fictional; code, experiment scripts, result tables and figure generators are in the project repository (MIT licence). © 2026 Csaba Bakay.

Table 12. Full comparison matrix (author's qualitative assessment, not a measurement): ratings per success factor and approach on the scale 1–5, factor model v0.5. "Excel" denotes a simple, manually maintained spreadsheet scorecard. Superscripts give the confidence: ^{H} verified from code or primary documentation, ^{M} plausible and partly documented, ^{L} assumption. For SaaS, the value is the mean of the known Prewave (P) and Sphera (S) values; the per-vendor values are shown in brackets where they differ or carry a mark. Marks: "?" unknown (no public evidence; not rated low and left out of the mean); "≥" documented minimum (only the publicly documented part is rated); "*" uncertain (kept at the earlier level because the evidence neither confirms nor contradicts the anchor). Example: 4 (P ≥4/S 4*) reads "Prewave at least 4 on documented evidence, Sphera 4 but uncertain". Generated from `research/comparison-matrix.yaml`; the rationale for every cell is in `research/comparison-matrix.md`.

#	Success factor	Excel	SaaS	OSS	Lean
1	Structured criteria & weights (objective + subjective)	3 ^{M}	4 ^{M}	2 ^{H}	4 ^{H}
2	Explainability & traceability (per result)	3 ^{M}	3 (P ≥3/S ?) ^{M}	2.5 ^{H}	4.5 ^{H}
3	Methodological transparency	4 ^{M}	2.5 (P 3/S 2) ^{M}	3 ^{H}	5 ^{H}
4	Expert calibration / ability to learn	1 ^{M}	2 ^{L}	1 ^{H}	4 ^{H}
5	Action framework	2 ^{M}	4 ^{M}	2 ^{H}	3.5 ^{H}
6	Human review & correction	2 ^{M}	4 (P ≥4/S 4*) ^{L}	2 ^{H}	4 ^{H}
7	External signals & data coverage	1 ^{H}	5 ^{M}	2 ^{H}	2.5 ^{M}
8	Regulatory evidence (due-diligence documentation)	2 ^{M}	4.5 (P 5/S 4) ^{M}	1 ^{H}	2 ^{H}
9	Cost / total cost of ownership	3 ^{M}	2 ^{M}	4 ^{M}	4 ^{M}
10	Time-to-value / ease of adoption	3 ^{M}	3 ^{L}	2 ^{H}	4 ^{M}
11	Customisability & data sovereignty	4 ^{M}	2 ^{M}	3 ^{H}	5 ^{H}
12	Scalability	2 ^{M}	5 ^{M}	2 ^{H}	3 ^{L}
13	Agentic orchestration & extensibility	1 ^{H}	2.5 (P 3/S 2) ^{M}	4 ^{H}	4 ^{H}
14	Operational maturity & support	3 ^{M}	5 ^{M}	1 ^{H}	2 ^{H}
15	Integration of risk management and performance evaluation	2 ^{M}	2 ^{L}	2 ^{H}	4 ^{H}